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# <Assignment 3>
# <Description>: In this program we will build up a game

import random
def printTitleMaterial():
    """ This function prints the 'title material' that prints out when the program
starts.
    """
    print("Spinner Winner!")
    print()
    print("By: <Pulkit Saxena>")
    print("[COM S 127 Section J]")
    print()

def initialChoice():
    """ This function allows the player to make various choices when starting the
game. This is an
    example of the 'do-while' pattern.

    :return String: The choice the player has made when starting the program to
[p]lay the game, view the
[i]nstructions, or [q]uit the program..
    """
    choice = input("Choice? [p]lay, [i]nstructions, [q]uit: ")
    while choice != "p" and choice != "i" and choice != "q":
        print("ERROR: Please enter 'p', 'i', or 'q'...")
        choice = input("Choice? [p]lay, [i]nstructions, [q]uit: ")
    return choice

def chooseNumPlayers():
    """ This function allows the player to choose whether they will play against
another
    human, or play against the computer.

    :return Integer: The number of players in the game.
    """
    numPlayers = 0
    while True:
        a = int(input("Are you playing againsta human or a computer\n1:
Computer\n2:Human\n"))
        if a==1:
            numPlayers+=1
            break
        if a==2:
            numPlayers+=2
            break
        else:
            print("Please enter 1 or 2")
    return numPlayers

def wait():
    """ This function has the computer 'wait' until the [Enter] key is pressed.
This allows
    for better 'readability' in the final output.
    """
    input("Press [Enter] To Continue...")
    print()

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def printBanner():
    """ Prints the 'banner' between each round of the game so that the output text
    does not
        get too 'messy.'
    """

print("#####")
print()
print("-- Starting New Round --")
print()

def printPoints(playerNum, points):
    """ This function prints the number of points a certain player currently has.

    :param Integer playerNum: The player whose points are being displayed.
    :param Integer points: The number of points to be displayed
    """
    print("* Player {0} Has {1} Points!".format(playerNum, points))
    print()

def wagerPointsHuman(playerNum, points):
    """ This function uses the 'do-while' pattern to take in input for the number
    of points to be wagered. It checks to
        make sure that the player has entered a valid amount of points. Meaning, the
        player cannot wager more points than
        they have, nor zero points, nor a negative number of points. The function then
        returns the wager.

    As there could potentially be two human players, this function requires the
    playerNum to know which player to include
        in any printouts.

    :param Integer playerNum: The player to include in any printouts. Can be 1 or
2.
    :param Integer points: The number of points the specified player currently has.
    :return Integer: The number of points to be wagered.
    """
    wager = 0
    while True:
        wager = (input("Player {0}, enter the points of the
wager".format(playerNum)))
        if 0<int(wager)<=int(points):
            break
        else:
            print("Please enter a positive number less than or equal to the amount
of points you have")
            print("* player {0} Has {1} Points!".format(playerNum,wager))
    return wager

def wagerPointsAI(playerNum, points):
    """ This function should choose a random number between 1 and the number of
    points the AI has (inclusive). For example,
        if the computer has 5 points, it can wager either 1, 2, 3, 4, or 5. This
    function should include a printout similar to
        what is printed when the human wagers points.

    :param Integer playerNum: The player to include in any printouts. While this
    will usually be 2, including this value allows
        for a future version of the game with 2 computer

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players.
:param Integer points: The number of points the specified player currently has.
:return Integer: The number of points to be wagered.
"""
    comwager = 0
    while True:
        comwager=random.randrange(1,(points+1))
        print("* player {0} Has wagered {1} Points!".format(playerNum,points))
        return comwager

def generateTargetValue(numSpinners, spinnerLow, spinnerHigh):
    """ This function generates the 'target value' that the players try to match.
    This number is generated by summing together
        random values between 'spinnerLow' and 'spinnerHigh' (inclusive), produced by a
    random number of spinners, between 1 and
        numSpinners (inclusive). For example, if there are 3 spinners, and a spinner
    can have values between 1 - 3, then the target
        value would be the summation of 1-3 random values between 1 and 3
    (inclusive).

    :param Integer numSpinners: The number of spinners used in the game.
    :param Integer spinnerLow: The smallest value a spinner can generate.
    :param Integer spinnerHigh: The highest value a spinner can generate.
    :return Integer: The target value generated by summing together 'numSpinners'
    number of random numbers between 'spinnerLow' and
        'spinnerHigh' (inclusive).
    """
    target = 0
    randNumspin=random.randint(1,numSpinners+1)
    for i in range (randNumspin):
        target+=random.randint(spinnerLow,spinnerHigh)
    return target

def getSpinnerChoiceHuman(playerNum, target, numSpinners, spinnerLow, spinnerHigh):
    """ This function gets the number of spinners that the human wants to spin. It
    should print out the 'target value' that the player
        is trying to match, as well as the values that a spinner can produce (ex: 1 -
    3), and the number of spinners that can be spun. The
        player cannot pick more spinners than are in the game, nor can they pick zero
    spinners, nor can they pick a negative number of spinners.

    :param Integer playerNum: The player to include in any printouts. Can be 1 or
    2.
    :param Integer target: The 'target value' the player is trying to match.
    :param Integer numSpinners: The total number of spinners in the game.
    :param Integer spinnerLow: The smallest value a spinner can generate.
    :param Integer spinnerHigh: The highest value a spinner can generate.
    :return Integer: The number of spinners the player chooses to spin.
    """
    spinnerChoice = 0
    while True:
        spinnerChoice=int(input("How many spinners do you want to spin?"))
        if spinnerChoice<= numSpinners and spinnerChoice>0:
            break
        elif spinnerChoice>numSpinners or spinnerChoice<0:
            print("Please select another number")
            print("* Player {0} can spin {1} points up to {2} points. Target
    value: {3}".format(playerNum, spinnerLow, spinnerHigh, target))

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return spinnerChoice
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def getSpinnerChoiceAI(playerNum, target, numSpinners, spinnerLow, spinnerHigh):  
    """ This function gets the number of spinners that the computer wants to spin.  
    This number should be a randomly generated value  
    between 1 and numSpinners (inclusive). It should print out text similar to what  
    the 'getSpinnerChoiceHuman()' function produces.
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        :param Integer playerNum: The player to include in any printouts. While this  
will usually be 2, including this value allows  
                                for a future version of the game with 2 computer  
players.
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        :param Integer target: The 'target value' the computer is trying to match. The  
computer does not take this value into account when  
                                choosing the number of spinners - it should be used for  
printouts, however.
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```
        :param Integer numSpinners: The total number of spinners in the game.  
        :param Integer spinnerLow: The smallest value a spinner can generate.  
        :param Integer spinnerHigh: The highest value a spinner can generate.  
        :return Integer: The number of spinners the computer chooses to spin.  
    """
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    spinnerChoice = 0  
    spinnerChoice=random.randrange(1,(numSpinners+1))  
    print("* Player {0} can spin {1} points up to {2} points. Target value:  
{3}").format(playerNum, spinnerLow, spinnerHigh, target))  
    return spinnerChoice
```

```
def spinSpinners(playerNum, spinnerChoice, target, spinnerLow, spinnerHigh):  
    """ This function can be used for either human or computer players, and it  
calculates the summed values of the number of  
    spinner spins. For example, if the player chooses to spin 3 spinners, and these  
spinners can have values between 1 and 3,  
    the player could spin values of 2, 3, and 1 for a total of 6. This is the value  
the function would return.
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    This function should print out the results of each spin as each spin is spun.  
The function should then print the sum of  
    all the spins and the target value once all the spins are complete.
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    Please note - the winner of the round is *not* calculated here - only the  
spinner totals.
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```
        :param Integer playerNum: The player to include in any printouts. Can be 1 or  
2.
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```
        :param Integer spinnerChoice: The number of spinners the player wishes to spin.  
        :param Integer target: Use this value in the printout so the user can compare  
what they spun compared to the target.
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        :param Integer spinnerLow: The smallest value a spinner can generate.  
        :param Integer spinnerHigh: The highest value a spinner can generate.  
        :return Integer: The sum of all the spinner spins.  
    """
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    spinVal = 0 # Assign the output of a random number between spinnerLow and  
spinnerHigh (inclusive) to this variable.
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    spinTotal = 0 # Sum the spinVal values together with this variable.
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    for i in range(spinnerChoice):
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        spinVal=random.randint(spinnerLow,spinnerHigh)
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        spinTotal+=spinVal
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        print(f"Player{playerNum} total spin value:{spinVal}")
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print(f"Player{playerNum} total spin value: {spinTotal}")
print(f"Target Value:{target}")

return spinTotal

def main():
    """ This is the main function that executes when the game is started from the
    terminal. It contains all of the logic/ states
    necessary to play the game.
    """
    # main script running control variable
    running = True

    # gameplay variables
    SPINNER_LOW = 1
    SPINNER_HIGH = 3
    NUM_SPINNERS = 3
    INITIAL_POINTS = 10
    player1Points = INITIAL_POINTS
    player2Points = INITIAL_POINTS

    # print the title/ author information
    printTitleMaterial()

    # play the game
    while running:
        choice = initialChoice()
        if choice == "p":

            numplayers = chooseNumPlayers()

            # main game loop
            while True:
                # round setup
                printBanner()

                target = str(generateTargetValue(NUM_SPINNERS, SPINNER_LOW,
                SPINNER_HIGH))

                # Player 1 wager. (Player 1 will always be human.)
                playerOneNum=1
                # choi=chooseNumPlayers()
                playerOneWager=wagerPointsHuman(playerOneNum,player1Points)
                print("Player One Wager is: "+ str(playerOneWager))
                # Player 2 wager. (Player 2 can be either human or AI - you must
                account for both.)
                playerTwoNum=2
                if numplayers==1:
                    playerTwoWager=wagerPointsAI(playerTwoNum,player2Points)
                    print("Player Two Wager is: "+ str(playerTwoWager))

                elif numplayers == 2:
                    playerTwoWager = wagerPointsHuman(playerTwoNum, player2Points)
                    print("Player Two Wager is: "+ str(playerTwoWager))

                # Player 1 spin - get the total of all the spinners. (Player 1 will
                always be human.)

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        spinnerChoiceOne = getSpinnerChoiceHuman(playerOneNum,
target, NUM_SPINNERS, SPINNER_LOW, SPINNER_HIGH)
        playerOneTotalPoints = spinSpinners(playerOneNum,
spinnerChoiceOne, target, SPINNER_LOW, SPINNER_HIGH)

        # Player 2 spin - get the total of all the spinners. (Player 2 can
be either human or AI - you must account for both.)
        if numplayers == 1:
            spinnerChoiceTwo = getSpinnerChoiceAI(playerTwoNum, target,
NUM_SPINNERS, SPINNER_LOW, SPINNER_HIGH)

        else:
            spinnerChoiceTwo = getSpinnerChoiceHuman(playerTwoNum, target,
NUM_SPINNERS, SPINNER_LOW, SPINNER_HIGH)

        playerTwoTotalPoints = spinSpinners(playerTwoNum, spinnerChoiceTwo,
target, SPINNER_LOW, SPINNER_HIGH)
        # Calculate Winner of the round. (If Player1 is closer, Player 1
wins. Else if Player 2 is closer Player 2 wins. If they are equal it is a draw.)
        if(int(target)- int(playerOneTotalPoints)>int(target)-
int(playerTwoTotalPoints)):
            print("Player 2 has won")
        elif(int(target)-int(playerOneTotalPoints)<int(target)-
int(playerTwoTotalPoints)):
            print("Player 1 has won")
        else:
            print("Tie")
        print("Player 1 had"+str(playerOneTotalPoints)+"points")
        print("Player 2 had"+str(playerTwoTotalPoints)+"points")
        print("The round has ended")
        if running==False:
            print("Game Over")
            playerOneTotalPoints=0
            playerTwoTotalPoints=0
            break

    elif choice == "i":
        print("The user selects a game between one or two players. ; A one
player game has the user play against the computer. ; A two player game has two
users play against one another. l The players start the game with a certain number
of 'points.' l The game is divided into 'rounds.' ; At the start of the 'round,'
the computer generates a 'target value.' n This is the value that the players try
to match. ; At the start of the 'round,' each player 'wagers' a certain number of
'points.' n Players cannot 'wager' more 'points' than they have. Nor can they
'wager' zero (0) 'points.' Nor can they 'wager' a negative number of 'points.'
; Then each player decides how many 'spinners' to spin. n Each spin adds its value
to a final 'spin value' for the round. n Players can not spin more spinners than
are available to be spun. Nor can the spin zero (0) spinners. Nor can they spin a
negative number of spinners. ; After each player spins their 'spinners,' their
'spin value' is compared against the 'target value.' ; The player who gets their
'spin value' closest to the 'target value' is the winner of the round. n If both
players 'spin value' are equally distant from the 'target value' the round is a
'draw.' ; Once the winner is decided, 'points' are added and subtracted from each
player's score. n The winning player gets their wager amount added to their score.
n The losing player gets their wager amount subtracted from their score. ; The game
continues until one player is completely out of 'points.'")

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        pass # Remove this 'pass' statement when other code has been added.
    elif choice == "q":
        print("GoodBye")
        Running=False
    else:
        pass # Remove this 'pass' statement when other code has been added.
        print("ERROR: Variable 'choice' should have been 'p', 'i', or 'q', but
instead was:", choice)
        quit()

        print("ERROR: Variable 'choice' should have been 'p', 'i', or 'q', but
instead was:", choice)
        quit()

if __name__ == "__main__":
    main()
```